Algebra: Chapter 0, Aluffi
Part II: Groups, First Encounter

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§3 The Category Grp

It must be shown that a unique group homomorphism $\sigma : A \times B \to G$ exists in the following commuting diagram:

3.

$$\begin{tikzcd}
 & & A \arrow[dl, "\iota_A"] \arrow[dll, bend right, "f"'] \\
G & A \times B \arrow[l, "\sigma"] & \\
 & & B \arrow[ul, "\iota_B"] \arrow[ull, bend left, "g"]
\end{tikzcd}$$

where $\iota_A : A \to A \times B$ is the natural inclusion map $a \mapsto (a, 0)$, and $\iota_B : B \to A \times B$; $b \mapsto (0, b)$. Then the fact that the diagram commutes and σ must be a group homomorphism forces the unique definition

$$\sigma(a, b) = f(a) + g(b).$$

Defined thus, $\sigma : A \times B \to G$ is indeed a group homomorphism because G is an abelian group:

$$\begin{aligned}
\sigma(a_1, b_1) + \sigma(a_2, b_2) &= f(a_1) + g(b_1) + f(a_2) + g(a_2) \\
&= f(a_1) + f(a_2) + g(a_1) + g(a_2) \\
&= \sigma(a_1 + a_2, b_1 + b_2).
\end{aligned}$$

This shows that $A \times B$ satisfies the universal property for the coproduct of A and B in Ab.

5.

Suppose $\mathbb{Q} \cong A \times B$ for some nontrivial groups A and B. Then there exist the natural projection maps $\pi_A : \mathbb{Q} \to A$ and $\pi_B : \mathbb{Q} \to B$. Notice that for natural projection maps from the product group to its component, say A, $(a, b) \in \ker \pi_A \iff a = 0$. Since $\mathbb{Q}$ is isomorphic to the product group,

$$\ker \pi_A \cap \ker \pi_B = 0 \tag{1}$$

Now, consider the two commuting diagrams from $\mathbb{Z}$ decomposing through $\mathbb{Q}$ to A and B:

$$\begin{tikzcd}
 & & a \in A \setminus 0 \\
1 \arrow[rr, mapsto, "!\sigma_1"] \arrow[urr, bend left] \arrow[drr, bend right] & & r_1 \in \mathbb{Q} \arrow[ur, mapsto, "\pi_A"] \arrow[dr, mapsto, "\pi_B"'] \\
 & & 0 \in B
\end{tikzcd}
\qquad
\begin{tikzcd}
 & & 0 \in A \\
1 \arrow[rr, mapsto, "!\sigma_1"] \arrow[urr, bend left] \arrow[drr, bend right] & & r_2 \in \mathbb{Q} \arrow[ur, mapsto, "\pi_A"] \arrow[dr, mapsto, "\pi_B"'] \\
 & & b \in B \setminus 0
\end{tikzcd}$$

Since $\pi_A(r_1) \neq 0$ and $\pi_B(r_2) \neq 0$, $r_1, r_2 \neq 0$ in $\mathbb{Q}$. By properties of the natural numbers, there are integers n, m such that $mr_1 = nr_2$. Applying π_A to both sides of this equation,

$$\pi_A(mr_1) = n0 = 0,$$

hence $mr_1 \in \ker \pi_A$. But $\pi_B(r_1) = 0 \implies mr_1 \in \ker \pi_B$ as well. Finally, since $r_1 \neq 0$, $mr_1 \neq 0$. This contradicts the fact that $\ker \pi_A \cap \ker \pi_B$ is trivial, as was established in (1).

6.

Suppose $Z_2 \times Z_3$ is the coproduct of Z_2 and Z_3 in Grp. Define the injective homomorphisms

$$f : Z_2 \to \mathfrak{S}_3; 1 \mapsto (\mathbf{1\,2})$$
$$g : Z_3 \to \mathfrak{S}_3; 1 \mapsto (\mathbf{1\,2\,3}).$$

Then by the universal property, there is a unique homomorphism $\sigma Z_2 \times Z_3 \to \mathfrak{S}_3$ such that the diagram below commutes:

$$\begin{tikzcd}
 & & Z_2 \arrow[dl, "\iota_2"] \arrow[dll, bend right, "f"'] \\
\mathfrak{S}_3 & Z_2 \times Z_3 \arrow[l, "\sigma"] & \\
 & & Z_3 \arrow[ul, "\iota_3"] \arrow[ull, bend left, "g"]
\end{tikzcd}$$

Namely, $(\mathbf{1\,2}), (\mathbf{1\,2\,3}) \in \sigma(Z_2 \times Z_3)$ because the diagram commutes. $(\mathbf{1\,2})$ and $(\mathbf{1\,2\,3})$ generate $\mathfrak{S}_3$, so σ is surjective. $Z_2 \times Z_3$ and $\mathfrak{S}_3$ are both finite groups, so σ is an isomorphism. By exercise 4.9, $Z_2 \times Z_3 \cong Z_6$, so it must be that $\mathfrak{S}_3$ has an element of order 6. This is not true, however; hence $Z_2 \times Z_3$ is not the coproduct of Z_2 and Z_3 in Grp.

8.

There are injections $\iota_2 : Z_2 \hookrightarrow G$; $1_2 \mapsto y$ and $\iota_3 : Z_3 \hookrightarrow G$; $1_3 \mapsto x$. Suppose there is a group H and homomorphisms

$$\varphi : Z_2 \to H; 1_2 \mapsto h_2$$
$$\psi : Z_3 \to H; 1_3 \mapsto h_3$$

$$\begin{tikzcd}
 & & Z_2 \arrow[dl, "\iota_2"] \arrow[dll, bend right, "\varphi"'] \\
H & G \arrow[l, "\sigma"] & \\
 & & Z_3 \arrow[ul, "\iota_3"'] \arrow[ull, bend left, "\psi"]
\end{tikzcd}$$

In the commuting diagram above, it must be shown that a unique homomorphism $\sigma : G \to H$ exists. Assuming a homomorphism σ exists, then mapping 1_2 from Z_2 gives $\sigma(y) = h_2$, and mapping 1_3 from Z_3 gives $\sigma(x) = h_3$. Because x and y generate G, these two mappings fully determine the hypothetical homomorphism σ. Indeed, such a homomorphism $\sigma : G \to H$ does exist: given each word $w = (w_1, \ldots, w_n) \in G$, map the letter

$$w_i \mapsto \begin{cases} h_2 & \text{if } w_i = y \\ h_3 & \text{if } w_i = x. \end{cases}$$

One can check that this is a group homomorphism, and furthermore $\sigma x = h_3$ and $\sigma(y) = h_2$.

9.

The fiber product exists in Grp. Given

$$\begin{tikzcd}
G \arrow[dr, "\varphi"] & \\
 & K \\
H \arrow[ur, "\psi"] &
\end{tikzcd}$$

$G \times_K H := \{(g, h) \in G \times H \mid \varphi(g) = \psi(h)\}$ is the fiber product of G and H with respect to K in Ab, just like the fiber product in Set. $G \times_K H$ is a subgroup of $G \times H$: given $(g_1, h_1), (g_2, h_2) \in G \times_K H$, the fact that φ and ψ are group homomorphisms means

$$\varphi(g_1 g_2) = \varphi(g_1)\varphi(g_2) = \psi(h_1)\psi(h_2) = \psi(h_1 h_2),$$

so $(g_1 g_2, h_1 h_2) \in G \times_K H$. Next, multiplying $\varphi(g_1^{-1})$ to the equation $\varphi(g_1) = \psi(h_1)$ yields

$$1 = \psi(h_1)\varphi(g^{-1}),$$

hence $\psi(h_1^{-1}) = \varphi(g^{-1})$. This shows the inverse of (g_1, h_1), namely (g_1^{-1}, h_1^{-1}), is in $G \times_K H$. Finally, of course $(1, 1) \in G \times_K H$. $G \times_K H$ is a subgroup of $G \times H$, so $G \times_K H$ is an object in Grp. Next, consider the commuting diagram

$$\begin{tikzcd}
 & & G \arrow[dr, "\varphi"] & \\
L \arrow[r, "\sigma"] \arrow[urr, bend left, "f"] \arrow[drr, bend right, "g"'] & G \times_K H \arrow[ur, "\pi_G"] \arrow[dr, "\pi_H"] & & K \\
 & & H \arrow[ur, "\psi"'] &
\end{tikzcd}$$

It must be shown that a unique homomorphism $\sigma : L \to G \times_K H$ exists. Indeed, the commutativity of the diagram forces $\sigma(l) = (f(l), g(l))$, and one can check that σ essentially inherits the homomorphism property from the fact that f and g are homomorphisms.

§5 Free Groups

6.

Recall that $F\{x\} \cong \mathbb{Z}$. By the universal property of the free group, there is a (unique) map $\iota_1 : \mathbb{Z} \to F\{x, y\}$ such that $1 \mapsto x$.

$$\begin{tikzcd} \mathbb{Z} \arrow[r, "\iota_1"] & F\{x,y\} \\ \{x\} \arrow[u] \arrow[ur] & \end{tikzcd} \qquad \begin{tikzcd} 1 \arrow[r, mapsto, "\iota_1"] & x \\ x \arrow[u, mapsto] \arrow[ur, mapsto] & \end{tikzcd}$$

Similarly, there is a map $\iota_2 : \mathbb{Z} \to F\{x, y\}$; $1 \mapsto y$. Take a group G and homomorphisms

$$\varphi : \mathbb{Z} \to G; 1 \mapsto g_1$$
$$\psi : \mathbb{Z} \to G; 1 \mapsto g_2$$

For the diagram

$$\begin{tikzcd} & & \mathbb{Z} \arrow[dl, "\iota_1"] \arrow[dll, bend right, "\varphi"'] \\ G & F\{x,y\} \arrow[l, "\sigma"] & \\ & & \mathbb{Z} \arrow[ul, "\iota_2"] \arrow[ull, bend left, "\psi"] \end{tikzcd}$$

to commute, a unique group homomorphism $\sigma : F\{x, y\} \to G$ sending $\sigma(x) = g_1$ and $\sigma(y) = g_2$ must exist. Indeed, such a σ is uniquely given by the universal property of the free group on $\{x, y\}$ with respect to $f : \{x, y\} \to G$; $x \mapsto g_1$, $y \mapsto g_2$.

$$\begin{tikzcd} F\{x,y\} \arrow[r, "!\sigma"] & G \\ \{x,y\} \arrow[u] \arrow[ur, "f"'] & \end{tikzcd}$$

This shows that $F\{x, y\}$ satisfies the universal property for the coproduct of $\mathbb{Z}$ and $\mathbb{Z}$ in Grp, hence $\mathbb{Z} * \mathbb{Z} \cong F\{x, y\}$.

8.

I only demonstrate the isomorphism for the category Grp since the argument for Ab is exactly the same. By the inclusion $A \hookrightarrow A \sqcup B \hookrightarrow F(A \sqcup B)$ and the universal property of the free group, there is a map $\iota_1 : F(A) \to F(A \sqcup B)$.

$$\begin{tikzcd} F(A) \arrow[r, "\iota_1"] & F(A \sqcup B) \\ A \arrow[u] \arrow[ur] & \end{tikzcd}$$

Similarly, there is a map $\iota_2 : F(B) \to F(A \sqcup B)$. To prove $F(A \sqcup B) \cong F(A) * F(B)$, it must be shown that given a group G and group homomorphisms $\varphi : F(A) \to G$ and $\psi : F(B) \to G$, there is a unique gropu homomorphism σ that makes the following diagram commute: Notice that, by the universal property

$$\begin{tikzcd} & & F(A) \arrow[dl, "\iota_1"] \arrow[dll, bend right, "\varphi"'] \\ G & F(A \sqcup B) \arrow[l, "\sigma"] & \\ & & F(B) \arrow[ul, "\iota_2"] \arrow[ull, bend left, "\psi"] \end{tikzcd}$$

Diagram 1

of the free group on A respectively, the entire homomorphism φ is determined uniquely by the restricted map $\varphi : A \to G$: letting $f := \varphi|_A$,

$$\begin{tikzcd} F(A) \arrow[r, "\varphi"] & G \\ A \arrow[u] \arrow[ur, "f"'] & \end{tikzcd}$$

Similarly, letting $g := \psi|_B$, ψ is determined uniquely by g. This means that if σ is indeed a homomorphism that makes diagram 1 commute, then

$$\sigma|_A = f$$
$$\sigma|_B = g$$

Indeed, the universal property for the coproduct of A and B combined the with the universal property for the free product of $A \sqcup B$ guarantees a unique σ that makes the diagrams below commute:

$$\begin{tikzcd} & & A \arrow[dl] \arrow[dll, bend right, "f"'] \\ G & A \sqcup B \arrow[l, "\sigma_0"] \arrow[dl] & \\ F(A \sqcup B) \arrow[u, "\sigma"] & & B \arrow[ul] \arrow[ull, bend left, "g"] \end{tikzcd}$$

where the unlabeled maps are just the respective inclusion maps and $\sigma_0 : A \sqcup B \to G$ is the unique function furnished by the universal property satisfied by $A \sqcup B$ with respect to f and g; σ is the unique homomorphism furnished by the universal property satisfied by $F(A \sqcup B)$ as a free group on $A \sqcup B$ with respect to σ_0. Since σ_0 is unique, σ is unique as well. It follows that σ is the unique homomorphism that makes diagram 1 commute.

10.

(a)

Suppose $\#A$ is finite. Then $F \cong \oplus_{i=1}^n \mathbb{Z}$ by question 7. Let $A = \{x_1, \ldots, x_n\}$; in F, identify x_i as $(0, \ldots, 0, 1, 0, \ldots, 0)$, where the 1 occurs in the ith position. It follows equivalence class of x_i under $\sim$ consists exactly of all $z \in \oplus_{i=1}^n \mathbb{Z}$ such that the ith coordinate is odd and all other coordinates are even. As such, the distinct elements of $F/\sim$ map bijectively with subsets of A, giving

$$\#(F/\sim) = 2^{\#A} < \infty.$$

Conversely, if $\#A$ is infinite, then the identification $F \cong \mathbb{Z}^{\oplus A}$ demonstrates that for all pairs $x \neq y \in A$, $\bar{x}$ and $\bar{y}$ are distinct equivalence classes in $F/\sim$. It follows that $\#F$ is infinite as well.

(b)

I'm not sure how to show this extra rigorously, but the intuition is clear: Since $F^{ab}(B)$ and $F^{ab}(A)$ are isomorphic groups, an equivalence relation $\sim_A$ on $F^{ab}(A)$ defined as in part (a) defines an analogous equivalence relation $\sim_B$ on $F^{ab}(B)$. Hence

$$F^{ab}(A)/\sim_A \cong F^{ab}(B)/\sim_B .$$

The desired results follow from this isomorphism of sets.

§6 Subgroups

8.

Suppose G is finitely generated, say by $A = \{x_1, \ldots, x_n\}$. Since A is finite, $F^{ab}(A) \cong \bigoplus_{i=1}^n \mathbb{Z}$. Consider the commuting diagram below:

$$\begin{tikzcd} \bigoplus_{i=1}^n \mathbb{Z} \arrow[r, "\varphi"] & G \\ A \arrow[u] \arrow[ur, hook, "\iota"] & \end{tikzcd}$$

Since $\iota(A) = A \subset G$, the image $\varphi(\bigoplus_{i=1}^n \mathbb{Z})$ includes a copy of A in G too. But A generates G and φ is a group homomorphism, so $\varphi(\bigoplus_{i=1}^n \mathbb{Z}) = G$ necessarily; φ is surjective.

Conversely, suppose there is a surjective homomorphism $\varphi : \bigoplus_{i=1}^n \mathbb{Z} \twoheadrightarrow G$. $\bigoplus_{i=1}^n \mathbb{Z} \cong \mathbb{Z}^n$ in Ab; the latter is generated by the standard bases, so let $e_1, \ldots, e_n$ analogously be the generators for $\bigoplus_{i=1}^n \mathbb{Z}$. Denoting $x_i = \varphi(e_i)$ for $i \in 1..n$, the fact that φ is a surjective homomorphism and that the e_i's generate $\bigoplus_{i=1}^n \mathbb{Z}$ means the x_i's generate A; A is thus finitely generated.

10.

The hint described by the textbook is essentially the extended Euclidean algorithm. After playing around with the matrices a bit, one can see that

- $t^n = \begin{bmatrix} 1 & n \\ & 1 \end{bmatrix}$, so $t^{-n} = \begin{bmatrix} 1 & -n \\ & 1 \end{bmatrix}$.
- $s^{-1}t^{-n}s = \begin{bmatrix} 1 & \\ n & 1 \end{bmatrix}$, so $s^{-1}t^n s = \begin{bmatrix} 1 & \\ -n & 1 \end{bmatrix}$.

Hence for all $n \in \mathbb{Z}$, $\begin{bmatrix} 1 & -n \\ & 1 \end{bmatrix}, \begin{bmatrix} 1 & \\ -n & 1 \end{bmatrix} \in H$. Take any $m = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Since m is invertible, $c = d = 0$ is impossible. Suppose $c = 0$. Then $\det m = 1$ means $ad = 1 \implies a = d = 1$ or $a = d = -1$. Either way, $m \in H$ since t^n, t^{-n}, and $s^2 = -I_2$ give the two desired forms for m; $m \in H$. Next, if $d = 0$, then $bc = -1$, so $m = \begin{bmatrix} a & 1 \\ -1 & \end{bmatrix}$ or $m = \begin{bmatrix} a & -1 \\ 1 & \end{bmatrix}$. Notice $t^n s = \begin{bmatrix} n & -1 \\ 1 & \end{bmatrix}$. It follows that in both cases, $m \in H$ once again.

Finally, suppose $c, d \neq 0$. Then, as the hint says,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & -q \\ & 1 \end{bmatrix} = \begin{bmatrix} a & -aq + b \\ c & -cq + d \end{bmatrix}$$
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & \\ -q & 1 \end{bmatrix} = \begin{bmatrix} a - bq & b \\ c - dq & d \end{bmatrix}$$

Without loss of generality, assume $|c| < |d|$. The Euclidean algorithm for the integers says there is an integer q such that $|d - cq| < |c|$. Hence multiply $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & -q \\ & 1 \end{bmatrix} = \begin{bmatrix} a & b_1 \\ c_1 & d_1 \end{bmatrix}$, where the specific value of b_1 is irrelevant, $c_1 = c$, and $|d_1| < |c_1|$ by choice of q. Now, continue performing the division algorithm for c_1 and d_1, etc., until one of the d_k's divide the corresponding c_k or vice versa; this state is guaranteed by the Euclidean algorithm in $\mathbb{Z}$. This leads to a reduction where $c_k = 0$ or $d_k = 0$, and hence this resulting matrix is in H. Since only matrices in H were multiplied to m to obtain the resulting matrix in H, it follows $m \in H$ as well. This proves that s and t generate $\mathrm{SL}_2(\mathbb{Z})$.

11.

Suppose $\mathfrak{S}_3 \cong Z_n * Z_m$ for some integers n and m. Consider the diagram

$$\begin{tikzcd}
 & & Z_n \arrow[dl, "j_1"] \arrow[dll, bend right, "\iota_1"'] \\
Z_n \times Z_m & \mathfrak{S}_3 \arrow[l, "\sigma"] & \\
 & & Z_m \arrow[ul, "j_2"] \arrow[ull, bend left, "\iota_2"]
\end{tikzcd}$$

where ι_1 and ι_2 are the respective inclusion maps into $Z_n \times Z_m$. Denote $e_1 := \iota_1([1]_n) = ([1]_n, 0)$ and $e_2 := \iota_2([1]_m) = (0, [1]_m)$. Also, let $w_1 := j_1([1]_n)$ and $w_2 := j_2([1]_m)$. Since the diagram commutes, $\sigma(w_1) = e_1$ and $\sigma(w_2) = e_2$. Because j_1, ι_1, j_2, ι_2 are all homomorphisms,

$$o(w_1) \mid o([1]_n) = n; o(e_1) = n \mid o(w_1)$$
$$o(w_2) \mid o([1]_m) = m; o(e_2) = m \mid o(w_2).$$

These relations show that $o(w_1) = n$ and $o(w_2) = m$. Recall that elements of $\mathfrak{S}_3$ have orders 1, 2, and 3. Namely, it wouldn't make sense for $o(w_2)$ or $o(w_3)$ to be 1 because that would imply either j_1 or j_2 are the zero maps, in which case $\mathfrak{S}_3$ doesn't satisfy the universal property for the coproduct of Z_n and Z_m. Hence $n, m \in \{2, 3\}$. First, consider the caes that $n \neq m$; that is, without loss of generality, assume $n = 2$ and $m = 3$. Then w_1 must be a two-cycle and w_2 a three-cycle. Furthermore, $\sigma(w_1 w_2) = \sigma(w_3) = ([1]_2, [1]_3)$ in $Z_2 \times Z_3$, where $w_2 = w_1 w_2$ is a two-cycle (the product of a three-cycle with a two-cycle in $\mathfrak{S}_3$ is a two-cycle). It follows that

$$0 = \sigma(w_3^2) = ([1]_2, [1]_3) + ([1]_2, [1]_3) = (0, [2]_3)$$

in $Z_2 \times Z_3$, which is clearly a contradiction.

Similarly contradictions can be derived for $(n, m) = (2, 2)$ and $(n, m) = (3, 3)$.

§7 Quotient Groups

4.

Suppose $f' \sim f$. It must be shown that for all $a \in F$,

$$af' \sim af \text{ and } f'a \sim fa.$$

Clearly this is so: by definition, $\exists g \in F$ such that $f - f' = 2g$, from which follows $af - af' = fa - fa' = 2ag$ with $ag \in F$.

Recall from §5, Q10 that equivalence classes under $\sim$ are analogous to subsets of A; in other words, $F/\sim$ is the group of words in A where each letter occurs at most once. In other words, $F/\sim \cong Z_2^{\oplus A}$.

11.

Take any $aba^{-1}b^{-1} \in [G, G]$ and $g \in G$. Denote $h = aba^{-1}b^{-1}$. Then $gaba^{-1}b^{-1}g^{-1} = ghg^{-1}$ and by definition of $[G, G]$, $ghg^{-1}h^{-1} \in [G, G]$. Since $[G, G]$ is a subgroup of G and $h \in [G, G]$,

$$ghg^{-1}h^{-1}h = gaba^{-1}b^{-1}g^{-1} \in [G, G].$$

This shows that $[G, G]$ is normal.

Take any $\bar{a}, \bar{b} \in G/[G, G]$. Since

$$\bar{a}\bar{b}\bar{a}^{-1}\bar{b}^{-1} = \bar{1}$$

in the quotient group, $\bar{a}\bar{b} = \bar{b}\bar{a}$, which shows that $G/[G, G]$ is commutative.

12.

By the universal property satisfied by the free group, f induces a unique group homomorphism $\varphi : F \to G$. Take any $a, b \in F$. Since φ is a group homomorphism and G is an abelian group, $\varphi(ab) = \varphi(a)\varphi(b) = \varphi(b)\varphi(a) = \varphi(ba)$, i.e.,

$$\varphi(aba^{-1}b^{-1}) \in \ker \varphi.$$

It follows that $[F, F] \subset \ker \varphi$. By the universal property satisfied by the quotient group, there is a unique group homomorphism $\bar{\varphi} : [F, F] \to G$ induced by φ. Since φ is uniquely induced by f, it can be said that f uniquely induces $\bar{\varphi}$; that is, $\bar{\varphi}$ is the unique group homomorphism that makes the diagram below commute:

$$\begin{tikzcd} F/[F,F] \arrow[r, "\bar{\varphi}"] & G \\ A \arrow[u] \arrow[ur, "f"'] & \end{tikzcd}$$

This is to say that $F/[F, F]$ satisfies the universal property for the free group of A in Ab. It follows that $F/[F, F] \cong F^{ab}(A)$.

Intuitively, it makes sense that $F/[F, F]$ is the free product over A in Ab: if $W = (w_1 w_2 \dots w_n) \in F(A)$ is a word in A, one can reduce W modulo $[F, F]$ to move each letter w_i anywhere in the word since $F/[F, F]$ is abelian. So in $F/[F, F]$, W can be reorganized as a finite sum of powers of elements of A, which is exactly the structure of $F^{ab}(A)$.

§8 Canonical Decomposition and Lagrange's Theorem

1.

$G_1 \cong G_2$ is not guaranteed. Take $G_1 = Z_6$ and $G_2 = \mathfrak{S}_3$. Both have Z_3 as a normal subgroup: $\langle [2]_6 \rangle \subset Z_6$ is isomorphic to Z_3 and normal in Z_6 because Z_6 is an abelian group. $\langle (\mathbf{1\,2\,3}) \rangle$ is isomorphic to Z_3 and normal in $\mathfrak{S}_3$ due to the fact that $\langle (\mathbf{1\,2\,3}) \rangle$ contains all the 3-cycles of $\mathfrak{S}_3$ and that two permutations are of the same type if and only if they are conjugates (not yet covered in the book). Hence $Z_6/Z_3 \cong \mathfrak{S}_3/Z_3 \cong Z_2$, but $Z_6 \ncong \mathfrak{S}_3$.

2.

Take $g \in G$. If $g \in H$, then $gH = Hg = H$. If $g \notin H$, then $H = H \sqcup gH = H \sqcup Hg$ since G has index 2. Because these are disjoint unions, it follows that $gH = Hg$. So in either case, $gH = Hg$, and this holds for all $g \in G$. This is an equivalent condition to the normality of H in G.

3.

Let G be a finite group. Then G is finitely presented by $(G \mid \mathscr{R})$, where $\mathscr{R}$ consists of all words gg^{-1} where g is an element of G. It's intuitively clear that $\mathscr{R}$ generates the kernal of the surjective homomorphism $\varphi : F(G) \to G$.

7.

Denote $H := F(A \cup A')/ < R \cup R' >$ where R and R' are generated by $\mathscr{R}$ and $\mathscr{R}'$, respectively. Define $\iota_1 : G \to H$ as the unique map satisfying

$$\begin{tikzcd}
F(A) \arrow[r] \arrow[rr, bend left, "j"] \arrow[dr] & F(A \cup A') \arrow[r] & F(A \cup A')/\langle R \cup R' \rangle \\
& F(A)/R \arrow[ur, "j_1"'] &
\end{tikzcd}$$

which is guaranteed to exist since the kernal of the composed map j contains R Identifying G as $F(A)/R$, write elements of G as a coset wR for some $w \in F(A)$. Then j_1 maps

$$wR \mapsto w\langle R \cup R' \rangle.$$

Define $j_2 : G' \to H$ analogously. To verify that H is indeed the coproduct of G and G' in Grp, suppose a group K and homomorphisms $\bar{\varphi} : G \to K$, $\bar{\psi} : G' \to K$ are given. It must be shown that a unique $\bar{\sigma} : H \to K$ makes the following diagram commute:

$$\begin{tikzcd}
& & G \arrow[dl, "j_1"] \arrow[dll, bend right, "\bar{\varphi}"'] \\
K & H \arrow[l, "\bar{\sigma}"] & \\
& & G' \arrow[ul, "j_2"] \arrow[ull, bend left, "\bar{\psi}"]
\end{tikzcd}$$

Diagram 2

Composing $\bar{\varphi}$ with the projection $F(A) \to F(A)/R$ recovers the homomorphism $\varphi : F(A) \to K$ with $R \subset \ker \varphi$. Similarly, recover $\psi : F(A') \to K$ with $R' \subset \ker \psi$. As was seen in exercise 5.8., $F(A \cup A')$ is the coproduct of $F(A)$ and $F(A')$ in Grp, so there is a unique homomorphism $\sigma : F(A \cup A') \to K$ induced by φ and ψ:

$$\begin{tikzcd}
& & F(A) \arrow[dl] \arrow[dll, bend right, "\varphi"'] \\
K & F(A \cup A') \arrow[l, "\sigma"] & \\
& & F(A') \arrow[ul] \arrow[ull, bend left, "\psi"]
\end{tikzcd}$$

Since the diagram commutes, $R, R' \subset \ker \sigma$. Since $\ker \sigma$ is a (sub)group, the subgroup generated by R and R' is in $\ker \sigma$ too:

$$\langle R, R' \rangle \subset \ker \sigma.$$

It follows from the first isomorphism theorem that there is a unique homomorphism $\bar{\sigma} : F(A \cup A')/\langle R, R' \rangle \to K$ induced by σ. This $\bar{\sigma}$ makes diagram 2 commute: for example, given $a \in A$, identify a as the coset aR in $F(A)/R \cong G$, and suppose $aR \overset{\bar{\varphi}}{\mapsto} k \in K$. Then

$$\begin{aligned}\bar{\sigma} j_1(aR) &= \bar{\sigma}(a\langle R \cup R' \rangle) \\ &= \sigma a \\ &= \varphi(a) \\ &= \bar{\varphi}(a) \\ &= k\end{aligned}$$

where the second equality follows because $\bar{\sigma}$ is the map induced by σ, and the third equality follows because σ is induced by φ for elements of $F(A)$. So the diagram commutes. On the other hand, if $\bar{\sigma} : H \to K$ is any homomorphism that makes diagram 2 commute, then $\bar{\sigma}$ should be described entirely by how $\bar{\sigma}$ maps A and A' (technically, their projections) because they generate the quotient $F(A \cup A')/\langle R, R' \rangle$. So indeed, the specific $\bar{\sigma}$ induced by $\sigma : F(A \cup A') \to K$ is the unique homomorphism that makes the diagram commute.

14.

Let $m := o(g)$. Since $\gcd(n, k) = 1$ and $m \mid n$ by Lagrange, $\gcd(m, k) = 1$. Hence $am + bk = 1$ for some $a, b \in \mathbb{Z}$. Since m is the order of g in G, $g^{am} = 1$, from which follows

$$g^{1-bk} = 1 \implies g = (g^b)^k.$$

So indeed, every element $g \in G$ is a kth power.

18.

Suppose $g \neq h \in G$ such that $o(g) = o(h) = 2$. Then clearly $\{1, g, h, gh\}$ is a subgroup of G of order 4, hence $4 \mid \#G$. This contradicts that assumption that $\#G = 2n$ for n odd, i.e., $4 \nmid \#G$.

Clearly, the statement is not true for a general group. For example, $\mathfrak{S}_3$ has order $6 = 2 \cdot 3$, but has many elements of order 2, namely, any of the transpositions: $(\mathbf{1\,2}), (\mathbf{1\,3}), (\mathbf{2\,3})$.

19.

The answer is 'no.' Consider the dihedral group $D_{12} = (x, y \mid x^6, y^2, xyxy)$. $4 \mid \#D_{12} = 12$, but there is no element of order 4: If $g \in D_{12}$ is a reflection, then g necessarily has order 2. If g is a rotation, then $g = x^k$ for some $1 \leq k < 6$. $\langle x \rangle \cong \mathbb{Z}/6\mathbb{Z}$, so the order of x^k in $\langle x \rangle$ (as a subgroup of D_{12} is given by $\frac{6}{\gcd(6,k)}$. Namely, this shows that $o(x^k) \in \{1, 2, 3, 6\}$ in $\langle x \rangle$, hence the order of $g = x^k$ cannot be 4 in the entire group D_{12}. All elements of D_{12} are either the identity, rotation, or reflection, so I have shown that no element of D_{12} has order 12.

20.

I use elementary facts from arithmetic. Induct on d. The case $d = 2$ follows immediately from exercise 8.17 and the fact that 2 is prime. Pick $\#G > d > 2$ and suppose for all $m \leq d - 1$, there is a subgroup of G of

order m. It must be shown that there is a subgroup H of G of order d. If d is prime, then the result follows by exercise 8.17. Suppose d is composite. If d is not the power of a prime, then let $d = ab$ for relatively prime integers a, b (e.g., pick such a pair via the fundamental theorem of arithmetic), and pick elements of $g, h \in G$ with orders a and b, respectively. Then one can use Lagrange for the subgroup $\langle g, h\rangle$, the fact that $\gcd(a, b) = 1$, and that G is abelian to check that $\langle g, h\rangle$ has order $ab = m$.

Consider the case that $d = p^n$ for a prime p and exponent $n > 1$. By the inductive hypothesis, there is an element $h \in G$ of order p^{n-1}. Let $H = \langle h\rangle$ and consider G/H. Writing $\#G = p^n k$ for some k, Lagrange says $\#G/H = pk$. Hence there is an element $lH \in G/H$ with order p (via Theorem 8.17, since p is prime and G/H is abelian). Letting $L = \langle lH\rangle$ be the subgroup generated by the coset lH in G/H, the correspondence theorem (an extension of Prop 8.9 in the textbook) says $\pi^{-1}(L)$ is a subgroup of G with order $\#L \cdot \#H = p^n$ ($\pi : G \to G/H$ denotes the projection).

21.

It's straightforward to check that

$$hH \cap K \mapsto hK$$

is a well-defined bijective map. To obtain the formala, notice $HK = \cup_{h\in H} hK$. Recall that distinct cosets are mutually disjoint. This gives

$$\begin{aligned}\#HK &= (\#\text{ of distinct cosets } hK)\#K \\ &= [H : H \cap K]\#K \\ &= \frac{\#H\#K}{\#(H \cap K)}\end{aligned}$$

where the second equality follows from the bijection above and third equalty follows from Lagrange: $\#H = [H : H \cap K]\#(H \cap K)$, which one can use because $H \cap K$ is a subgroup of H.

22.

Consider the diagram

$$\begin{tikzcd} G \arrow[r, "\varphi"] \arrow[rr, bend left, "0"] & G' \arrow[r, "\alpha"] \arrow[d, "\pi"] & L \\ & G'/N \arrow[ur, "\exists!\bar{\alpha}"'] & \end{tikzcd}$$

Since $\alpha\varphi = 0$, $\operatorname{im}\varphi \subset \ker\alpha$. Additionally, $\ker\alpha$ is normal in G', so $N \subset \ker\alpha$ by definition of N. Hence the uniquely determined $\bar{\alpha} : G'/N \to L$ induced by α makes triangle block starting from G' commute, whereas $\operatorname{im}\varphi \subset N \subset \ker\alpha$ guarantees the diagram commutes when starting from G.

24.

Consider the projection $\pi : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$. Although it is surjective, and is hence an epimorphism in Grp (the direction (c) $\implies$ (a) does not use the assumption that φ is a homomorphism between abelian groups), π doesn't have a right inverse: the only homomorphism $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$ is the trivial homomorphism, and clearly the trivial homomorphism is not a right inverse to the projection.

25.

When H is normal in G, the "interesting" homomorphism $\mathrm{Inn}(G) \to \mathrm{Aut}(H)$ is the map $\gamma \mapsto \gamma|_H$. This map can be extended to $G \to \mathrm{Aut}(H)$ by composing

$$g \mapsto \gamma_g \mapsto \gamma_g|_H .$$

Suppose additionally that H is abelian. Then there is a map

$$\begin{aligned} G/H &\to \mathrm{Aut}(H) \\ gH &\mapsto \gamma_g|_H . \end{aligned} \tag{2}$$

I check that this map is well-defined; verifying that the map is indeed a group homomorphism is easy. Suppose $g_1H = g_2H$. It must be shown that $\gamma_{g_1} : H \to H$ and $\gamma_{g_2} : H \to H$ are equivalent homomorphisms. $g_2^{-1}g_1 \in H$, say $g_2^{-1}g_1 = h'$ for some $h' \in H$. Take any $h \in H$. Then

$$\begin{aligned} \gamma_{g_1}(h) &= g_1^{-1}hg_1 \\ &= g_2h'hh'^{-1}g_2^{-1} \\ &= g_2hg_2^{-1} = \gamma_{g_2}(h) \end{aligned}$$

where the third equality follows from the commutativity of H. Since γ_{g_1} and γ_{g_2} agree on every value of H, they are equivalent homomorphisms; $\gamma_{g_1} = \gamma_{g_2}$ in $\mathrm{Aut}(H)$. Hence the map given in (2) is well-defined.

§9 Group Actions

8.

Recall that $\mathsf{G\text{-}Set}$ consists of objects (ρ, A), where $\rho : G \times A \to A$ defines a group action on A, and morphisms from $(\rho, A) \to (\rho', A')$ are given by set functions $\varphi : A \to A'$ such that the diagram below commutes:

$$\begin{tikzcd} G \times A \arrow[r, "\mathrm{id}_G \times \varphi"] \arrow[d, "\rho"'] & G \times A' \arrow[d, "\rho'"] \\ A \arrow[r, "\varphi"'] & A' \end{tikzcd}$$

As was noted in the text, such φ are exactly equivariant functions. This construction is indeed a category. Obviously, id_A is G-equivariant, so $\mathcal{H}(A, A)$ has an identity; that id_A is indeed the identity map in $\mathcal{H}(A, A)$ is induced from id_A satisfying the condition for the identity map in Set. The associativity of the morphisms in $\mathsf{G\text{-}Set}$ follows from the associativity of morphisms in Set as well. Finally, morphisms in $\mathsf{G\text{-}Set}$ can be composed: if $\varphi : A \to B$ and $\psi : B \to C$ are equivariant, then the composed function $\psi\varphi$ is equivariant as well:

$$g\psi\varphi(a) = \psi(g\varphi(a)) = \psi\varphi(ga),$$

hence $\psi\varphi \in \mathcal{H}(A, C)$.

It is obvious that homomorphisms $\varphi : A \to A'$ in $\mathsf{G\text{-}Set}$ are equivariant bijections. Conversely, suppose $\varphi : A \to A'$ is an equivariant bijection. To show that φ is an isomorphism in $\mathsf{G\text{-}Set}$, it must be demonstrated that the inverse function $\varphi^{-1} : A' \to A$ is also in $\mathsf{G\text{-}Set}$, i.e., φ^{-1} is equivariant. Indeed, this is true because the diagram below commutes:

$$\begin{tikzcd} G \times A' \arrow[r, "\mathrm{id}_G \times \varphi^{-1}"] \arrow[d, "\rho'"'] & G \times A \arrow[d, "\rho"] \\ A' \arrow[r, "\varphi^{-1}"'] & A \end{tikzcd}$$

This is because

$$\begin{tikzcd} g \times \varphi(a) \arrow[r, mapsto] \arrow[d, mapsto] & g \times a \arrow[d, mapsto] \\ \underset{=\varphi(ga)}{g\varphi(a)} \arrow[r, mapsto] & ga \end{tikzcd}$$

for all $g \in G$, all $\varphi(a) \in A'$.

11.

I use slightly different notation than the book. The action of G on left cosets G/H by left multiplication can be realized as the induced permutation representation (not yet covered in the book...)

$$\begin{aligned} \Phi : G \to \mathrm{Aut}(G/H); \quad & g \mapsto \sigma_g, \\ \sigma_g : G/H \to G/H; \quad & aH \mapsto gaH \end{aligned}$$

Since $\#(G/H) = p$, $\mathrm{Aut}(G/H) \cong \mathfrak{S}_p$, so Φ can be understood as a homomorphism into $\mathfrak{S}_p$. By the first isomorphism theorem, $G/\ker\Phi \cong S < \mathfrak{S}_p$. Since S is a subgroup of $\mathfrak{S}_p$, $[G : \ker\Phi] = \#S \mid \#\mathfrak{S}_p$, i.e.,

$$[G : \ker\Phi] \mid p!. \tag{3}$$

Next, notice $\ker\Phi < H$: If $g \in \ker\Phi$, then $\sigma_g = \mathrm{id}_{G/H}$, which means $H = \sigma_g(H) = gH$, i.e., $g \in H$. By the tower law(?) for indices, $[G : \ker\Phi] = [G : H][H : \ker\Phi]$, i.e.,

$$[G : \ker\Phi] = p[H : \ker\Phi] \tag{4}$$

Combining constraints (3) and (4), $[H : \ker\Phi] \mid (p-1)!$. Namely, this shows that the prime factors of $[H : \ker\Phi]$ are less than p. Then it is necessary that $[H : \ker\Phi] = 1$; if not and a prime $q < p$ divides $[H : \ker\Phi]$, then $q \mid \#G$ by Lagrange, which contradicts the assumption that p is the smallest prime factor of $\#G$. $[H : \ker\Phi] = 1$ simply means $H = \ker\Phi$, hence H is be a normal subgroup of G.

13.

Suppose $f : G/H \to G/gHg^{-1}$ is an isomorphism between left multiplication by G on G/H and G/gHg^{-1} in $\mathsf{G\text{-}Set}$. Because f is equivariant, the mapping $H \mapsto x(gHg)^{-1}$, where $x \in G$ is a placeholder group element, determines all of f:

$$f(aH) = af(H) = ax(gHg^{-1})$$

Such a map must be well defined: if $aH = bH$, then by assumption that f is indeed well defined,

$$ax(gHg^{-1}) = bx(gHg^{-1}) \implies x^{-1}b^{-1}ax \in gHg^{-1},$$

which means $x = g^{-1}$ satisfies the desired properties. Indeed, one can determine that the map

$$f(aH) = ag^1(gHg^{-1})$$

determines an equivariant bijection $G/H \to G/gHg^{-1}$

17.

Let $\varphi : G \to G$ be an equivariant bijection that makes the following diagram commute:

$$\begin{tikzcd} G \times G \arrow[r, "\mathrm{id}\times\varphi"] \arrow[d, "l"'] & G \times G \arrow[d, "l"] \\ G \arrow[r, "\varphi"] & G \end{tikzcd}$$

where $l : G \times G \to G$ is the action $G \curvearrowright G$ by left multiplication. Since φ is equivariant, the mapping is entirely determined by $\varphi(1) = x$ for some $x \in G$:

$$\varphi(g) = g\varphi(1) = gx$$

So any equivariant bijection $G \to G$ can be identified by where 1 gets mapped to. Define

$$\begin{aligned} \Phi : G \to \mathrm{Aut}_{\mathsf{G\text{-}Set}}(G); &\quad g \mapsto \varphi_g, \\ \varphi_g : G \to G; &\quad h \mapsto hg \end{aligned}$$

Then one can verify that Φ is a homomorphism with inverse

$$\Psi : \mathrm{Aut}_{\mathsf{G\text{-}Set}}(G) \to G; \quad \varphi \mapsto \varphi(1),$$

hence $G \cong \mathrm{Aut}_{\mathsf{G\text{-}Set}}(G)$ as groups.

§10 Group Objects in Categories

1.

The only interesting assertions I want to check are that the compositions

$$G \xrightarrow{\epsilon\times\mathrm{id}} 1 \times G \xrightarrow{\pi} G$$

and

$$1 \times G \xrightarrow{\mathrm{id}} G \xrightarrow{\epsilon\times\mathrm{id}} 1 \times G$$

are both identities. To check that the first composition gives the identity, let $\psi : G \to H$ and $\varphi : K \to G$ be any morphisms. Then the diagrams below commute because of the universal property for products:

$$\begin{tikzcd} & 1 \times G \arrow[dr, "\pi"] & \\ G \arrow[ur, "\epsilon\times\mathrm{id}"] \arrow[rr, "\mathrm{id}"] \arrow[dr, "\psi"'] & & G \arrow[dl, "\psi"] \\ & H & \end{tikzcd} \qquad \begin{tikzcd} & 1 \times G \arrow[dr, "\pi"] & \\ G \arrow[ur, "\epsilon\times\mathrm{id}"] \arrow[rr, "id"] & & G \\ & K \arrow[ul, "\varphi"] \arrow[ur, "\varphi"'] & \end{tikzcd}$$

hence $\pi(\epsilon \times \mathrm{id})\psi = \psi$ and $\varphi\pi(\epsilon \times \mathrm{id}) = \varphi$, or in other words, $\pi(\epsilon \times \mathrm{id}) = \mathrm{id}$.

The second composition can be verified as the identity morphism once again by noting that the diagram below commutes:

$$\begin{tikzcd}
1 \times G \arrow[r, "\pi_G"] \arrow[rrr, bend left, "\pi_1"] \arrow[rrrr, bend right, "\pi_G"'] & G \arrow[r, "\epsilon \times \mathrm{id}"] & 1 \times G \arrow[rr, "\pi_1"] \arrow[rr, "\pi_G"'] & & 1 \\
& & & & G
\end{tikzcd}$$

Here, the top arrow $1 \times G \to 1$ is indeed π_1 because 1 is final; there is only one map $1 \times G \to 1$, and it is the projection. The bottom arrow is $1 \times G \to G$ is π_G because, by the composition identity shown above,

$$\pi_G(\epsilon \times \mathrm{id})\pi_G = (\pi_G(\epsilon \times \mathrm{id}))\pi_G = \mathrm{id}_G \pi_G = \pi_G$$

Therefore, the universal property for the product gives a map $\sigma : 1 \times G \to 1 \times G$ with repsect to π_1 and π_G, namely $\sigma = \mathrm{id}_{1\times G}$:

$$\begin{tikzcd}
1 \times G \arrow[r, "\pi_G"] \arrow[rrr, bend left, "\pi_1"] \arrow[rr, bend left, "\mathrm{id}_{1\times G}"] \arrow[rrrr, bend right, "\pi_G"'] & G \arrow[r, "\epsilon \times \mathrm{id}"] & 1 \times G \arrow[rr, "\pi_1"] \arrow[rr, "\pi_G"'] & & 1 \\
& & & & G
\end{tikzcd}$$

which verifies that $(\epsilon \times \mathrm{id})\pi_G = \mathrm{id}_{1\times G}$.